Siemens S700K Point Machine Switching Time Specifications

Operating Tolerances, Fault Conditions, and Environmental Factors

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1. Overview

Switching time is the elapsed time from the initiation of a switch command to the confirmation of the locked position (lock-to-lock time). It is measured by the S700K internal controller and logged to the CMMS telemetry system. Switching time is an important diagnostic indicator: values outside the normal range indicate either mechanical resistance (high time) or a locking mechanism fault (low time).

2. Normal Operating Range

The following values apply at ambient temperatures between -10 C and +40 C. Times are measured from command initiation to locking relay energisation.

Condition	Switching Time	Action
Normal operation	4.0 - 5.5 seconds	No action required
Elevated - monitor	5.5 - 6.4 seconds	Log and monitor trend over 7 days
High - inspect within 48h	6.5 seconds	Inspect for mechanical resistance
Critically high - withdraw	> 6.5 seconds	Withdraw from service; Priority 2 defect
Abnormally fast	< 3.5 seconds	Withdraw from service; possible locking fault - Priority 1

WARNING: A switching time below 3.5 seconds may indicate a failure of the locking mechanism. The switch blades may not be achieving full lock, creating a derailment risk. Withdraw the asset from service immediately and escalate to a Level-3 engineer.

3. Common Causes of Elevated Switching Time

- Gearbox bearing wear - typically presents as a gradual increase over weeks, correlated with rising motor current (cross-reference MAN-PM-002).
- Switch bed debris or fouling - sudden onset; inspect switch bed for ballast, vegetation, ice, or foreign objects.
- Drive rod misalignment or damage - check rod for bending or wear at joints.
- Hardened lubrication (winter) - cold grease increases friction; see MAN-PM-004.
- Blade obstruction - check for deformation or track geometry issues preventing full throw.

4. Switching Time Under Extreme Temperatures

The S700K is rated for ambient temperatures from -25 C to +70 C, but switching times outside the -10 C to +40 C normal range may exhibit the following behaviour:

Temperature Range	Expected Effect	Additional Action
Below -10 C	Up to 1.0 s increase is acceptable	Inspect lubrication pre-winter (MAN-PM-004)
-10 C to +40 C	Normal range applies	Standard monitoring
Above +40 C	Minor decrease possible	Inspect for thermal expansion of components if > 10%

NOTE: If switching time repeatedly exceeds 6.0 seconds in cold weather despite correct lubrication, consider installing a point heating system. Raise a capital works request through the infrastructure planning team.